

BoardNet: Board Examination Communication Network

Introduction

The Ministry of Education has decided to modernize the communication system used during the Higher Secondary Certificate (HSC) Board Examination across Bangladesh. To support secure coordination among examination offices, a unified digital network named BoardNet will be designed and implemented.

For simplicity, the network will include only six HSC education boards based on six districts of Bangladesh. These boards are responsible for handling exam coordination, result processing, administrative communication, and digital service access.

The estimated number of hosts/devices required for each board network is given below:

Examination Unit	Abbreviation	Estimated Hosts
Dhaka Board	DHK	300
Rajshahi Board	RAJ	160
Chattogram Board	CTG	200
Barishal Board	BAR	120
Sylhet Board	SYL	150
Khulna Board	KHU	120

Requirements

Topology Design & Server Setup

- **Core Nodes:** Every HSC board is represented by a router.
- **Isolated Regional Board:** The Chattogram Board, CTG, is only connected to the Dhaka Board, DHK. It has no direct connection to any other board router.
- **Central Examination Coordination Switch:** DHK, SYL, and RAJ are connected to one another using a central switch.
- **Direct Administrative Links:** DHK, KHU, and BAR are connected to one another directly using point-to-point router links. DHK-KHU, KHU-BAR, BAR-DHK.
- **Local Email Servers:** Every board has its own local Email Server for official examination communication. You must configure the email servers accordingly. The hostnames should follow the format: (mail.dhk.edu.bd, mail.ctg.edu.bd etc)
- **Dedicated Web Servers:**
 - Only DHK and CTG have dedicated web servers. (www.dhk.edu.bd and www.ctg.edu.bd)
 - You must configure the web servers so that every device from every board network can access both websites using domain names.
- **Central DNS Server:** Only DHK hosts the central DNS Server. You have to configure the DNS server so that all devices across all board networks can use it for domain name resolution.
- **Main DHCP Provider:** The DHK router acts as the DHCP server for all board networks except KHU and BAR.
- **Dedicated DHCP Servers:** KHU and BAR each have their own dedicated local DHCP Server. You must configure these DHCP services properly.
- **End Devices:** Each board has its own LAN switch and must have one printer, one email server, and exactly two PCs shown in the topology to represent all hosts/devices of that board.

Addressing

- Select the student ID of the first member of your group.
- Take the last 4 digits of that ID and divide them into two parts containing two digits each.
- These two parts will become the first two octets of the base network address.
- The third and fourth octets are 0, and the prefix mask is /16.
- Subnet this base network using VLSM to provide a proper network address to each HSC board network.
- All PCs in the board networks will receive IP addresses dynamically, while all other devices (Routers, Servers, Printers) must be configured manually

Example

Suppose the student ID is: 20201002

The last four digits are: 1002

Dividing it into two parts gives: 10 and 02

Therefore, the base network address is: **10.2.0.0/16**

DHCP Configuration

- DHK's router acts as the DHCP server for: DHK, CTG, SYL and RAJ
- KHU and BAR has their own dedicated local DHCP server for the Khulna and Barishal Board networks.
- Since CTG is connected only through DHK, CTG devices must receive their dynamic IP addresses through the DHK router.

Routing

- **Dynamic Routing:** DHK, SYL, and RAJ will share their routing tables with one another using the RIP protocol.
- **Full Awareness:** The DHK router must be aware of all possible networks in the topology, either dynamically or statically.
- **Default Gateway:** CTG can reach all other board networks only through DHK. Therefore, a default route must be configured on the CTG router.
- **Static Mapping:** KHU and BAR must be configured to know about all other networks using static routing.

- **Redundancy:** If the direct path between DHK and KHU goes down,, then KHU must send packets through BAR using recursive routing. Similarly, if the direct path between DHK and BAR goes down, then BAR must still reach the network through KHU.

You are allowed to make any valid and necessary networking assumptions while designing the network infrastructure, provided they do not conflict with or simplify any of the requirements mentioned above.

Testing & Verification

After completing the configuration, you must verify the following conditions:

- Every device across all six HSC board networks must be able to ping every other device.
- All PCs must receive IP addresses according to the DHCP requirements.
- Devices from every board must be able to access:
 - www.dhk.edu.bd
 - www.ctg.edu.bd
- DNS resolution must work from every board network.
- Email communication must be configured among all boards.
- You must demonstrate the redundancy conditions by shutting down the relevant primary interface or link:
 - When the DHK–KHU link goes down, KHU must still reach the network through BAR.
 - When the DHK–BAR link goes down, BAR must still reach the network through KHU.

Deliverables

1. **Work Distribution:** Clearly mention which group member configured which board, service, or routing section.
2. **Packet Tracer File:** Submit the completed .pkt / .pka file with all required devices and configurations.
3. **Network Topology Diagram:** Submit a picture of the full network topology diagram with proper labels. The diagram must include:
 - Board names
 - Router names
 - Server names
 - LAN segments
 - Inter-router links
 - Network addresses written as notes beside every network/link
4. **PDF Report:** Submit a PDF report containing:
 - Complete VLSM tree
 - Full IP addressing table (unit, device, IP address, subnet mask, default gateway)
 - The configuration commands of all the 6 routers you have implemented.
 - DNS configuration
 - DHCP configuration
 - Web server configuration details
 - Email server configuration details